Inferential Statistics (IS)

A Project Report

Submitted to

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By

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The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain.

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Problem 3:

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients.

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Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favor one method above another and may work better in his/her favorite method. The response is the variable of interest.

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Problem 1

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	Forward	Attacking Midfielder	Winger	**Total**
Players Injured	45	56	24	20	**145**
Players Not Injured	32	38	11	9	**90**
Total	**77**	**94**	**35**	**29**	**235**

1.1 What is the probability that a randomly chosen player would suffer an injury?

Solution:

To calculate the probability that a randomly chosen player would suffer an injury. We need to divide the number of injured players by the total number of players.

Number of injured players = 145

Total number of players = 235

Probability of a randomly chosen player suffering an injury

= Total Number of Players Injured/ Total Number of Players

= 145/235

= 0.617.

So, the probability that a randomly chosen player would suffer an injury is approximately 0.617.

1.2 What is the probability that a player is a forward or a winger?

Solution:

The total number of players in the forward position is 94, and the total number of players in the winger position is 29.
As these are mutually exclusive events, a player can be a forward or winger.
We can calculate the probability as:
Probability that a player is a forward or a winger
= (Number of Forwards + Number of Wingers) / Total Number of Players
= (94+29)/235
= 0.523.
Therefore, the probability that a player is a forward or a winger is approximately 0.523.

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

Solution:

To calculate this probability, we need to consider the number of players who play in the striker position and have a foot injury, divided by the total number of players.

Number of players in striker position = 77
Number of injured players who play in striker position = 45
Total number of players = 235

Probability that a randomly chosen player plays in a striker position and has a foot injury
= Number of Injured Strikers/ Total Number of Players
= 45/235
= 0.191.
Therefore, the probability that a randomly chosen player plays in a striker position and has a foot injury is approximately 0.191.

1.4 What is the probability that a randomly chosen injured player is a striker?

Solution:

To find the probability that a randomly chosen injured player is a striker, we need to divide the number of injured players in the striker position by the total number of injured players:

Probability that a randomly chosen injured player is a striker

= Number of Injured Strikers/ Total Number of Injured Players

= 45/145

= 0.31.

Therefore, the probability that a randomly chosen injured player is a striker is approximately 0.31.

Problem 2:

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain.

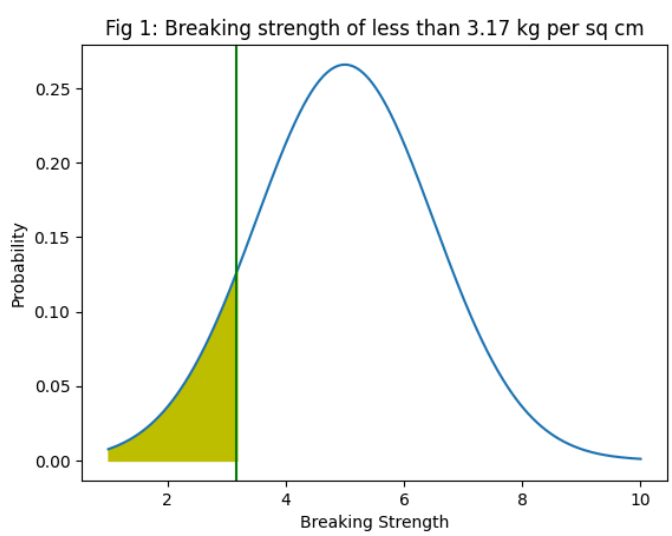
- Imported all the required packages.
- Here we will use the normal distribution
    1. cdf (cumulative density function) = cdf(x, mu(μ), sigma(σ))
    2. pdf (probability density function) = pdf(x, mu(μ), sigma(σ))
- Where, x = observed value, mu(μ) = mean, sigma(σ) = standard deviation
- mu = 5 kg per sq.centimeter and sigma = 1.5 kg per sq.centimeter

3.1 What proportion of the gunny bags have a breaking strength less than 3.17 kg per sq cm?

Solution:

```
probability_less_than_3_17 = norm.cdf(3.17, mu, sigma)
```

By looking at the below graph we can observe the proportion of gunny bags having a breaking strength less than 3.27 kg per sq. cm.is 0.1112.

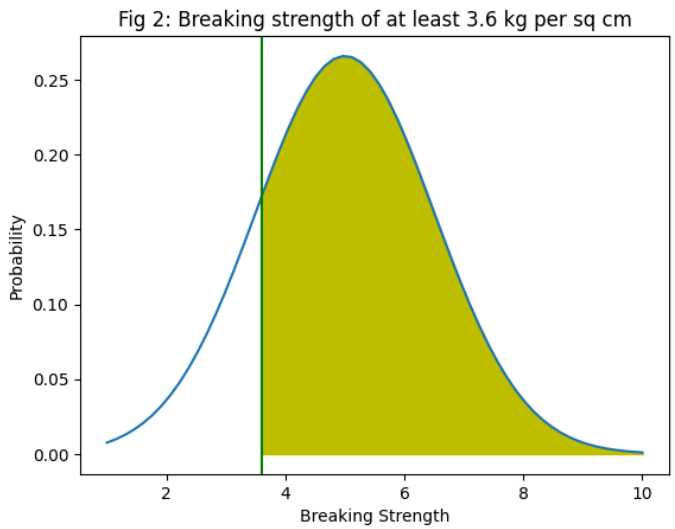


2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

Solution:

```
probability_more_than_3_6 = 1 - norm.cdf(3.6, mu, sigma)
```

By looking at the below graph we can easily observe that the proportion of gunny bags having a breaking strength at least 3.6 kg per sq. cm. is 0.8247.

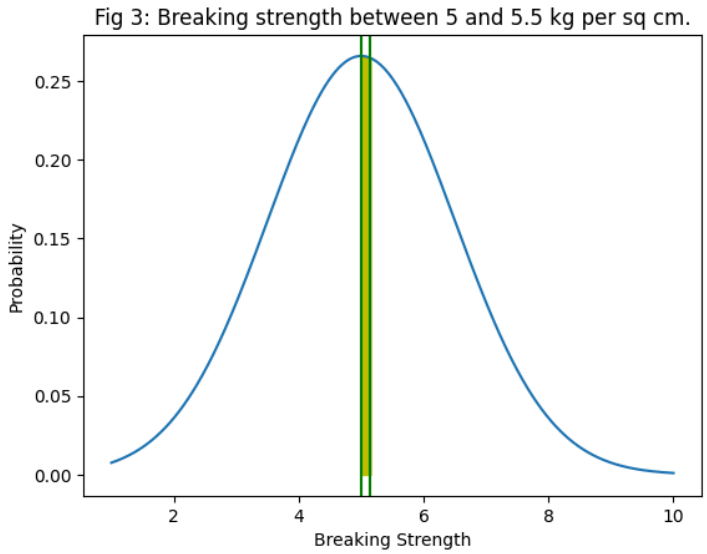


2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

Solution:

```
Probablity_between_5_and_5_5 =
        stats.norm.cdf(5.5,mu,sigma) -  stats.norm.cdf(5,mu,sigma)
```

From the below graph we can have a look at the proportion of the gunny bags having a breaking strength between 5 and 5.5 kg per sq cm =0.1306.

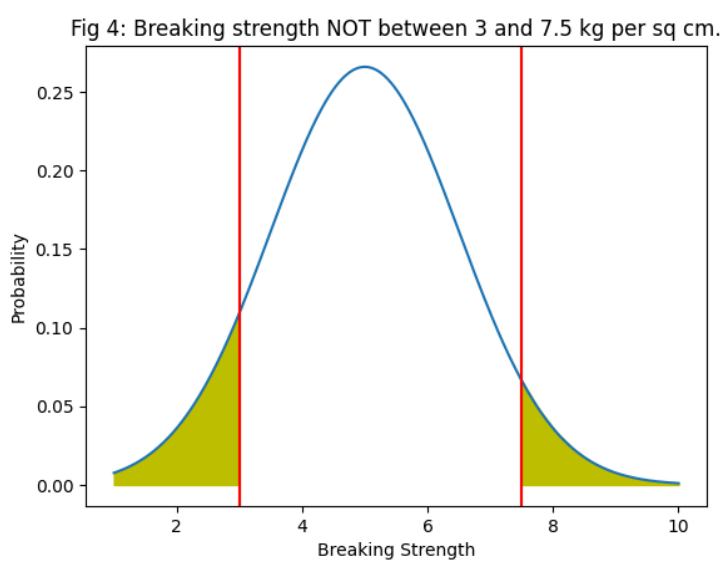


2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?

Solution:

```
probability_not_between_3_a_7_5 =
            1 - (stats.norm.cdf(7.5,mu,sigma) - stats.norm.cdf(3,mu,sigma))
```

By having a look at the below graph we can observe the proportion of the gunny bags having a breaking strength Not between 3 and 7.5 kg per sq cm. = 0.139.



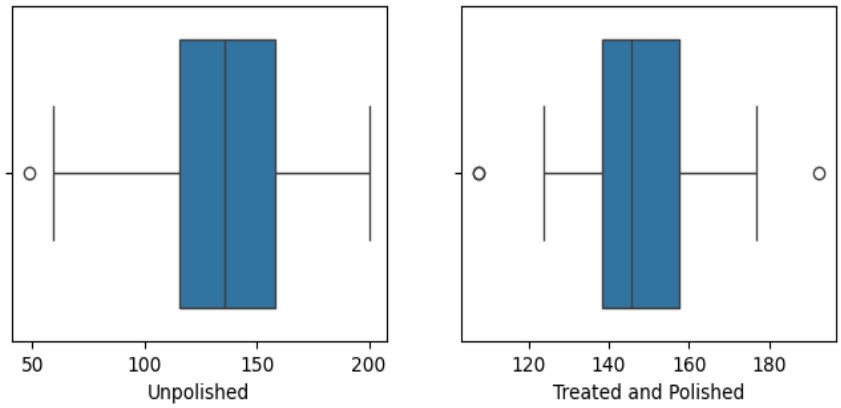
Problem 3:

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

1. Imported all the required libraries.
2. Loaded the zingaro company dataset using **read_csv().**
3. Used **head()** to get the first 5 rows of the data.
4. Checked the shape of the dataset using the **shape method.**
    a. Data set having 75 rows.
    b. Data set having 2 columns.
5. Checked the data types of all the columns in the dataset using **info().**
    a. All the columns are floating values.
    b. There are no missing values in the dataset.
6. Checked the statistical summary of the given data set using **describe().**
    a. The count of Unpolished stone and Treated and Polished are 75 each.
    b. The mean of Unpolished and Treated and Polished stones are 134.110527 and 147.788117 respectively.

Box Plots for Unpolished, Treated and Polished

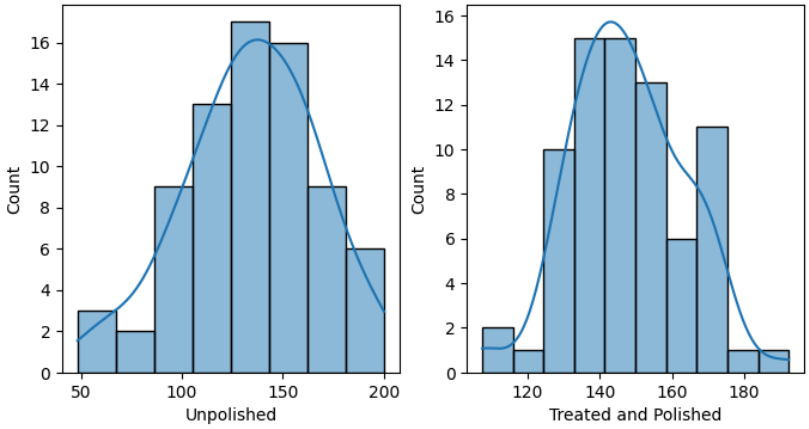
Fig 5: Box Plots for Unpolished, Treated and Polished



1. The mean of the two samples is not equal.
2. There are few outliers in both the data.

Histograms for Unpolished, Treated and Polished

Fig 6: Histograms for Unpolished, Treated and Polished



Histogram is depicting that data from each of the 2 groups following a normal distribution.

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

Solution:

We can take the below values in consideration to do the calculation

Level of Significance: α = 0.05

Sample Size = 75

Now, we will form the hypothesis.

Null Hypothesis: Unpolished stone is suitable for printing which means Brinell's hardness index is atleast or equal to 150.

Alternative Hypothesis: Unpolished stone is not suitable for printing which means Brinell's hardness index is less than 150.

- Ho: μ >= 150
- Ha: μ < 150
  Where μ be the population mean of unpolished stones.

Used the independent one sample one tailed t-test to calculate T_stats and p-value

```
t_stat, p_value = stats.ttest_1samp(df['Unpolished '], popmean = 150, alternative = 'less')
```

Conclusion

The p-value(4.171286997419652e-05) is less than the level of significance. so we can reject the null hypothesis with 95% confidence, we can say that we have enough evidence to say that the hardness for unpolished stones is less than 150.

Thus Zingaro is right in thinking that polished stones are not fit for painting.

3.2 Is the mean hardness of the polished and unpolished stones the same?

Solution:

We can take the below values in consideration to do the calculation

Level of Significance: α = 0.05

Sample Size = 75

Now, we will form the hypothesis.

Null Hypothesis: The mean hardness of the polished and unpolished stones is the same.

Alternative Hypothesis: The mean hardness of the polished and unpolished stones are not the same.

Ho: μ(Unpolished) = μ(Treated and polished)

Ha: μ(Unpolished) ≠ μ(Treated and polished)

Where μ(Unpolished) be the population mean of Unpolished stones and μ(Treated and polished) be the population mean of Treated and polished

The alternative hypothesis indicates that the test is two-tailed and here we will use two independent Sample tests.

```
test_stat, p_value = stats.ttest_ind(df['Unpolished '], df['Treated and Polished'],
equal_var = False, alternative = 'two-sided')
```

Conclusion

The p-value is less than the level of significance. so we can reject the null hypothesis with 95% confidence, we can say that there is insufficient evidence to say that the mean hardness of the polished and unpolished stones are the same.

Problem 4:

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favor one method above another and may work better in his/her favorite method. The response is the variable of interest.

1. Imported all the required libraries.
2. Loaded the “Dental Hardness data” dataset using **read_excel().**
3. Used **head()** to get the first 5 rows of the data.
4. Checked the shape of the dataset using the **shape method.**
    a. Data set having 90 rows.
    b. Data set having 5 columns.
5. Checked the data types of all the columns in the dataset using **info().**
    a. All the columns are integer values.
    b. There are no missing values in the dataset.
6. Changing Dentist, Method and Alloy data types to categorical using **Categorical()**. And re-checked the type of columns to ensure the changes were made on the dataset.
    a. The Dentist , Method and Alloy columns are categorical.
    b. Temp and Response columns are numerical.
7. Checked the statistical summary of the given data set using **describe().**
    a. There are 5 Unique values in the Dentist column.
    b. There are 3 Unique values in the Method column.
    c. There are 2 Unique values in the Alloy column.
8. Using the **value_count()** got the count of each unique value present in the Dentist, Method and Alloy column.
    a. Dentist column have 1, 2, 3, 4 and 5 as unique values each of 18 counts.

b. Method column have 1, 2, and 3 as unique values each of 30 counts.
c. Alloy column have 1 and 2 as unique values each of 45 counts.

9. Sample is equally divided among Dentists, Methods and Alloy Types.
10. Separated the data of Alloy Type 1 and Alloy Type 2 into two different variables. Checked the first 5 rows using **head()**, the shape using **shape()** and statistical summary using **describe()** for each.

4.1 How does the hardness of implants vary depending on dentists?

Solution:

Firstly, we will form the hypothesis,

Null and Alternative Hypothesis for the two types of alloys:

For Alloy Type1:

Null Hypothesis(Ho):There is no difference among the dentists on the implant hardness.

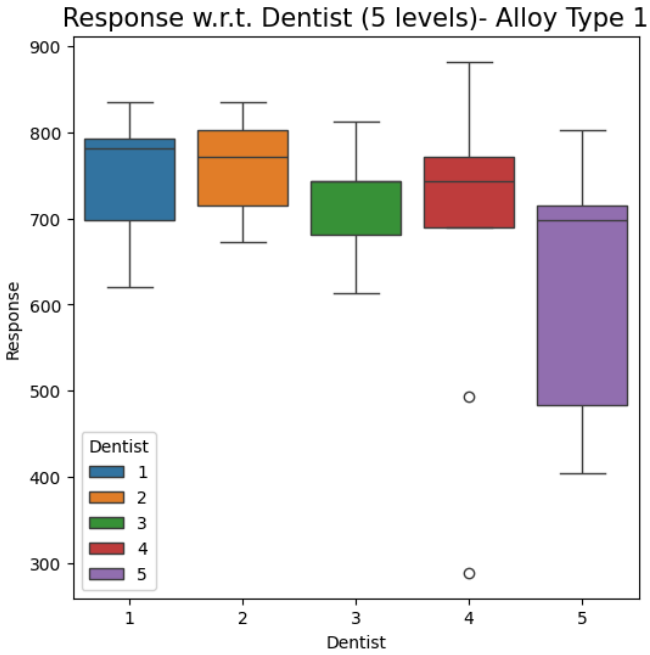
Alternative Hypothesis(Ha): There is a difference among the dentists on the implant hardness.

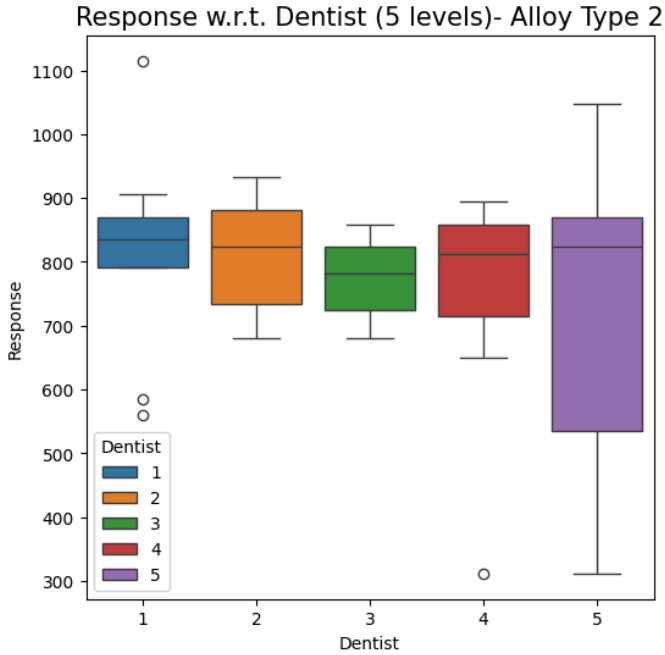
For Alloy Type2:

Null Hypothesis(Ho): There is no difference among the dentists on the implant hardness.

Alternative Hypothesis(Ha): There is a difference among the dentists on the implant hardness.

The Mean of Response at different levels of Dentist factor





- Tested two types of Alloys Separately using one-way Anova test, got the following conclusion.

The assumption of hypothesis testing.

For Alloy Type 1:

- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - Yes, the homogeneity of variance assumption is verified using the **Levene's test.**

For Alloy Type 2:

- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - Yes, the homogeneity of variance assumption is verified using the **Levene's test.**

The p-value for Alloy Type 1 and Alloy Type 2 Using one-way Anova test.

For Alloy Type 1: p = 0.116 → No significant difference among dentists.

For Alloy Type 2: p = 0.718 → No significant difference among dentists.

Conclusion:

For Alloy 1: p = 0.116 → No significant difference among dentists.

For Alloy 2: p = 0.718 → No significant difference among dentists.

- Therefore, dentists do not significantly affect the hardness of implants, regardless of the alloy used.
- Tukey's HSD were not performed since ANOVA did not show statistically significant differences among dentists for either alloy.

4.2 How does the hardness of implants vary depending on methods?

Solution:

Firstly, we will form the hypothesis,

Null and Alternative Hypothesis for the two types of alloys:

For Alloy Type1:

Null Hypothesis(Ho):There is no difference among the methods on the implant hardness.

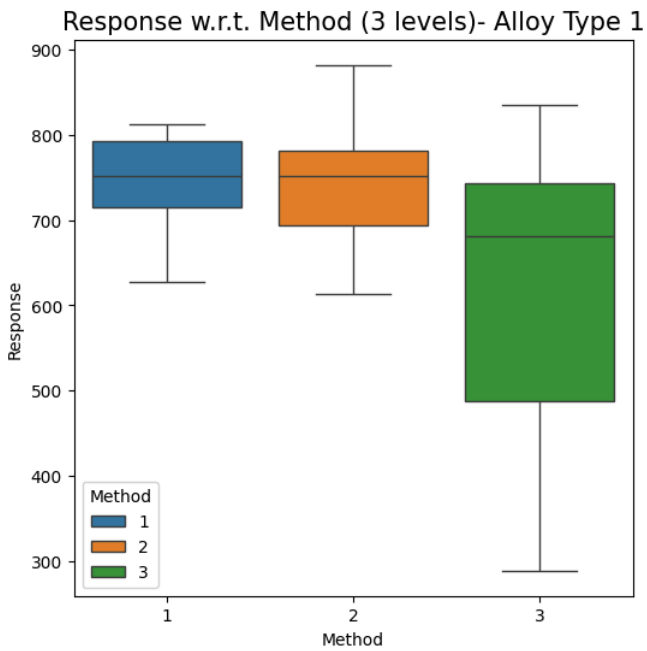
Alternative Hypothesis(Ha): There is a difference among the method on the implant hardness.

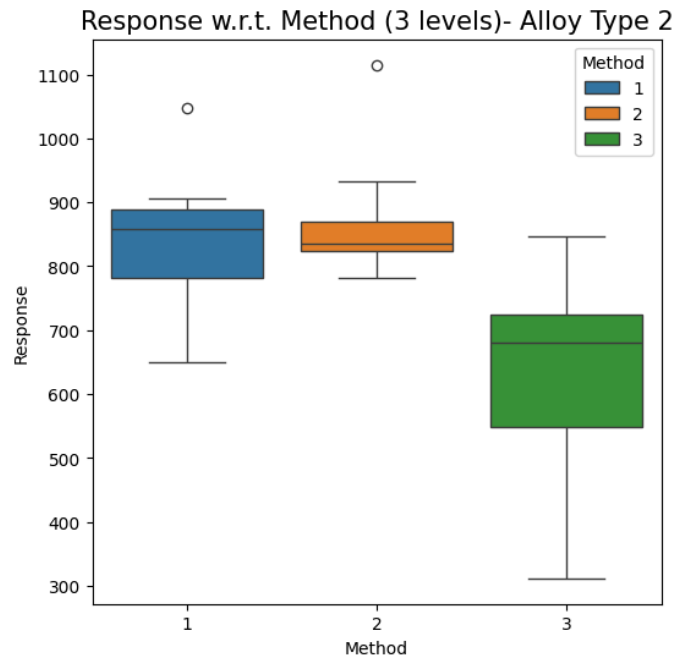
For Alloy Type2:

Null Hypothesis(Ho): There is no difference among the method on the implant hardness.

Alternative Hypothesis(Ha): There is a difference among the method on the implant hardness.

The Mean of Response at different levels of Method factor





- Tested two types of Alloys Separately, got the following conclusion.

The assumption of hypothesis testing.

For Alloy Type 1:

- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - No, the homogeneity of variance assumption is verified using the **Levene's test.**

For Alloy Type 2:

- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - No, the homogeneity of variance assumption is verified using the **Levene's test.**

The p-value for Alloy Type 1 and Alloy Type 2 Using one-way Anova test.:

For Alloy 1: p = 0.00416341216750554 24 → There is a significant difference among Method.

For Alloy 2: p = 5.4158710514431865e-06 → There is a significant difference among Method.

Note: Tukey's HSD were performed since ANOVA did show statistically significant differences among Method for either alloy.

Multiple Comparison test (Tukey HSD)

Alloy Type 1:

In order to identify for which method mean response is different from other groups,

Null hypothesis(H0):

μ1=μ2 and μ1=μ3 and μ2=μ3

Alternative hypothesis(Ha):

μ1≠μ2 or μ1≠μ3 or μ2≠μ3

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
1	2	-6.1333	0.987	-102.714	90.4473	False
1	3	-124.8	0.0085	-221.3807	-28.2193	True
2	3	-118.6667	0.0128	-215.2473	-22.086	True

As the p-values(refer to the p-adj column) for comparing the mean response, we can say that,

- Group 3 differs significantly from both Group 1 and Group 2.
- There is no significant difference between Group 1 and Group 2.
- This suggests that the method used in Group 3 impacts the response differently.

Multiple Comparison test (Tukey HSD)

Alloy Type 2:

In order to identify for which method mean response is different from other groups,

Null hypothesis(H0):

μ1=μ2 and μ1=μ3 and μ2=μ3

Alternative hypothesis(Ha):

μ1≠μ2 or μ1≠μ3 or μ2≠μ3

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
1	2	27.0	0.8212	-82.4546	136.4546	False
1	3	-208.8	0.0001	-318.2546	-99.3454	True
2	3	-235.8	0.0	-345.2546	-126.3454	True

As the p-values(refer to the p-adj column) for comparing the mean response, we can say that,

- There is no significant difference between Method 1 and Method 2.
- Method 3 is significantly different from both Method 1 and Method 2.
- This means Method 3 gives different results compared to the other two.

Conclusion:

For Alloy Type 1: p = 0.0041634121675055424 → There is a significant difference among Method.

For Alloy Type 2: p = 5.4158710514431865e-06 → There is a significant difference among Method.

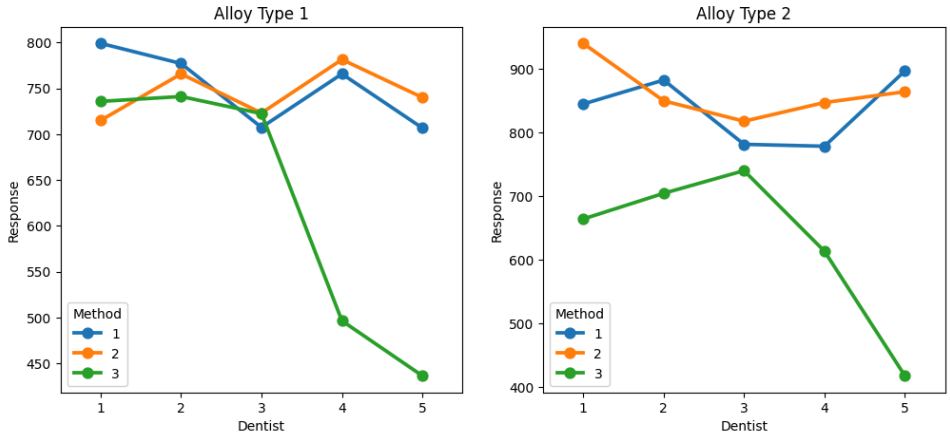
- Therefore, Method do have significantly affect the hardness of implants, regardless of the alloy used.
- Tukey's HSD for alloy 1 suggests that Method 3 impacts the response differently.

- Tukey's HSD for alloy 2 suggests that Method 3 gives different results compared to the other two.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

Solution:

Fig 7: Interaction Effect between the Method and Dentist on the hardness of dental implants for Alloys



Inferences from the Interaction Plots:

For Alloy Type 1:

- The hardness values vary between dentists, and this variation depends on the method.
- For some dentists, Method 1 gives higher hardness, while for others, Method 2 is better.
- The crossing lines between Method 1 and Method 2 indicate a clear interaction effect.

For Alloy Type 2:

- The crossing lines between Method 1 and Method 2 indicate a clear interaction effect.
- The difference in hardness between methods is not consistent across dentists, again confirming an interaction.
- Some dentists show a strong shift in performance depending on the method used.

Anova Table for Alloy Type 1:

	sum_sq	df	F	PR(>F)
C(Dentist)	106683.688889	4.0	3.899638	0.011484
C(Method)	148472.177778	2.0	10.854287	0.000284
C(Dentist):C(Method)	185941.377778	8.0	3.398383	0.006793
Residual	205180.000000	30.0	NaN	NaN

Inference for Alloy Type 1:

- Dentist effect is significant ($p \approx 0.0115$), there are differences between dentists.
- Method effect is highly significant ($p \approx 0.0003$), method strongly affects the response.
- Interaction effect is significant ($p \approx 0.0068$), the effect of method depends on which dentist applies it.

Anova Table for Alloy Type 2:

	sum_sq	df	F	PR(>F)
C(Dentist)	56797.911111	4.0	1.106152	0.371833
C(Method)	499640.400000	2.0	19.461218	0.000004
C(Dentist):C(Method)	197459.822222	8.0	1.922787	0.093234
Residual	385104.666667	30.0	NaN	NaN

Inference for Alloy Type 2:

- Dentist effect is not significant ($p \approx 0.3718$), dentists perform similarly.
- Method effect is highly significant ($p \approx 0.000004$), method has a strong impact on response.
- Interaction effect is not significant at 5% level ($p \approx 0.0932$), weak evidence that method performance depends on the dentist.

Conclusion:

- For Alloy Type 1, the interaction between Dentist and Method is significant ($p = 0.0068$), meaning that the effectiveness of a method depends on which dentist applies it. In other words, some dentists perform better with certain methods than others.
- For Alloy Type 2, the interaction between Dentist and Method is not significant ($p = 0.0932$), meaning that the effectiveness of the method is consistent across dentists. The choice of dentist does not influence how well a method works for Alloy Type 2.
- For Alloy Type 1, careful matching of dentist and method is important to optimize performance. For Alloy Type 2, method choice alone matters — dentist selection does not affect the outcome.

4.4 How does the hardness of implants vary depending on dentists and methods together?

Solution:

Null and Alternative Hypothesis for the two types of alloys:

For Alloy Type1:

Null Hypothesis(Ho):There is no difference among the dentists and method together on the implant hardness

Alternative Hypothesis(Ha): There is difference among the dentists and method together on the implant hardness

For Alloy Type2:

Null Hypothesis(Ho): There is no difference among the dentists and method together on the implant hardness

Alternative Hypothesis(Ha): There is difference among the dentists and method together on the implant hardness

The assumption of hypothesis testing.

For Alloy Type 1:

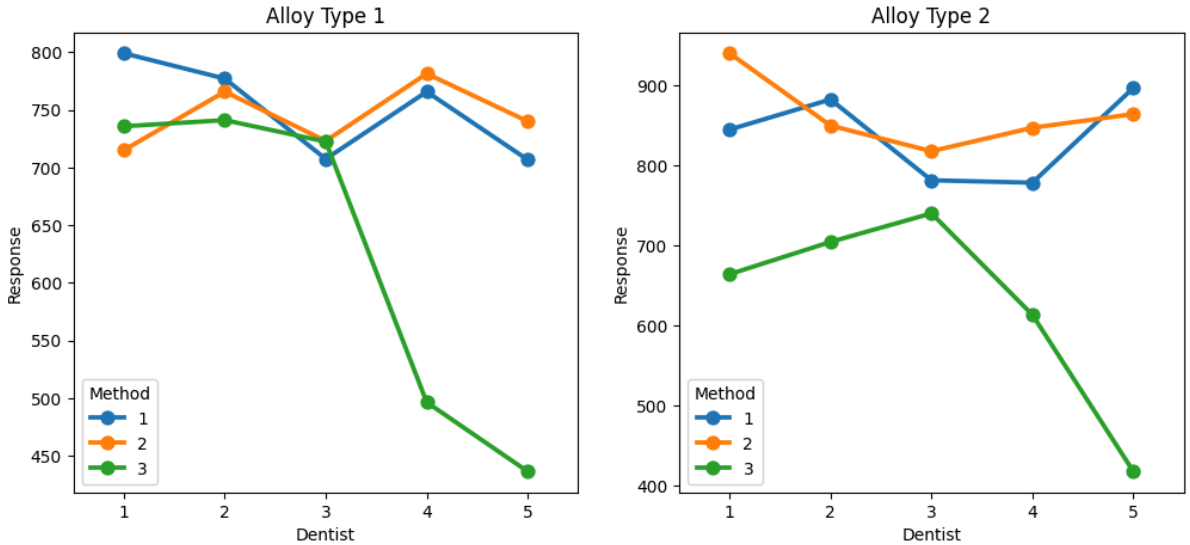
- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - No, the homogeneity of variance assumption is verified using the **Levene's test.**

For Alloy Type 2:

- The populations are normally distributed - No, the normality assumption is verified using the **Shapiro-Wilk's test.**
- Population variances are equal - No, the homogeneity of variance assumption is verified using the **Levene's test.**

Interaction Effect plot:

Fig 8: The hardness of implants vary depending on dentists and methods together for Alloys



For Alloy Type 1:

- The hardness values vary between dentists, and this variation depends on the method.
- For some dentists, Method 1 gives higher hardness, while for others, Method 2 is better.
- The crossing lines between Method 1 and Method 2 indicate a clear interaction effect.

For Alloy Type 2:

- The crossing lines between Method 1 and Method 2 indicate a clear interaction effect.
- The difference in hardness between methods is not consistent across dentists, again confirming an interaction.
- Some dentists show a strong shift in performance depending on the method used.

Calculate the p-value for Alloy Type 1 using anova table

	sum_sq	df	F	PR(>F)
C(Dentist)	106683.688889	4.0	3.899638	0.011484
C(Method)	148472.177778	2.0	10.854287	0.000284
C(Dentist):C(Method)	185941.377778	8.0	3.398383	0.006793
Residual	205180.000000	30.0	NaN	NaN

Inference for Alloy Type 1:

1. Dentist effect is significant ($p \approx 0.0115$), there are differences between dentists.
2. Method effect is highly significant ($p \approx 0.0003$), method strongly affects the response.
3. Interaction effect is significant ($p \approx 0.0068$), the effect of method depends on which dentist applies it.

Calculate the p-value for Alloy Type 2 using anova table

	sum_sq	df	F	PR(>F)
C(Dentist)	56797.911111	4.0	1.106152	0.371833
C(Method)	499640.400000	2.0	19.461218	0.000004
C(Dentist):C(Method)	197459.822222	8.0	1.922787	0.093234
Residual	385104.666667	30.0	NaN	NaN

Inference for Alloy Type 2:

1. Dentist effect is not significant ($p \approx 0.3718$), dentists perform similarly.
2. Method effect is highly significant ($p \approx 0.000004$), method has a strong impact on response.
3. Interaction effect is not significant at 5% level ($p \approx 0.0932$), weak evidence that method performance depends on the dentist.

Conclusion:

1. For Alloy Type 1, both dentist and method significantly affect response, and their interaction also matters.
2. For Alloy Type 2, method strongly affects response, but dentist differences and interaction are not significant.
3. Overall, method choice is consistently important across both datasets, while the dentist effect and interaction are more dataset-dependent.